

Emotion Detection in Social Media with Retrieval-Augmented Generation (RAG) Justification

Bachelor Semester Project 5

Student Name: SOUSA

FARIA João Bernardo

ID:0232421751

University Name: University of
Luxembourg

Email:

joao.sousa.003@student.uni.lu

Preferred Language: French

Tutor Name: LAMSIYAH

Salima

University Name: University of
Luxembourg

Email:

salima.lamsiyah@uni.lu

ABSTRACT

The project was developed by student **João Bernardo Sousa Faria**, under the supervision of their tutor **Salima Lamsiyah**. The project aims to build a system that detects emotions (e.g., anger, joy, sadness) in social media posts (tweets or Reddit comments) and explains its predictions. Instead of being a black box, the system will use Retrieval-Augmented Generation (RAG) to justify the detected emotion by retrieving and presenting similar posts from emotion-labeled datasets. For example, a post classified as *anger* will be accompanied by evidence such as: *"Similar posts expressing anger also use terms like 'furious' and 'disgusted'."*

Overview of the Scientific section of the project:

In the scientific section, the student studies the question **"How can RAG improve explainability and trust in emotion detection systems for social media?"**. The mentioned question focuses on learning about the purpose of RAG in its totality, as well as in the development of the system.

Overview of the Technical section of the project:

As previously mentioned, the project consists of building a system programmed to detect emotions in social media posts while explaining its predictions. The objective can be achieved with **Python** as the main programming language, with the help of libraries such as **NumPy** and **PyTorch**.

Additionally, at least one **dataset** is required for the development of the system. In this project, datasets such as

GoEmotions (Reddit), SemEval-2018 Affect in Tweets or other emotion-labeled corpora can be used to fill the role.

Expected Outputs

By the end of the project, The reader is expected to learn about:

- An emotion classification model;
- A retrieval module to provide supporting examples;
- An evaluation of both accuracy and explanation quality.

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REFERENCES